MTA AUDIO EDITOR

M-Live / Merish Karaoke MP3 Profile - Interoperability Specification

Karaoke MP3 format technical specification



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Build: 20261011140234

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Licenza: EUPL-1.2

Digital Audio Workstation with MTA support

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Purpose

This profile produces a normal stereo MP3 file with synchronized ID3v2.3 metadata for lyrics, chords and markers. The file remains playable by generic MP3 players while adding karaoke information intended for M-Live/Merish devices and other ID3/SYLT-compatible readers.

Compatibility baseline

Merish devices advertise MP3 lyric support through ID3/SYLT and provide editing/display of lyrics, chords and markers on MP3 backing tracks. MTA Audio Editor therefore uses standard ID3 primitives while keeping the MP3 audio stream independent from karaoke metadata.

File structure

The output consists of an ID3v2.3 tag at the beginning of the file followed by the MPEG Layer III audio stream. Any previous ID3v2 tag is replaced rather than nested. The audio stream is not re-encoded while karaoke metadata is inserted.

ID3 frames

Frame	Purpose
TIT2	Title
TPE1	Artist / performer
TCOM	Authors / composers
USLT	Complete unsynchronized lyrics
SYLT type 1	Synchronized lyrics
SYLT type 5	Synchronized chords
SYLT type 4	Synchronized markers / events
TXXX	Profile identification

SYLT synchronization

Each SYLT frame uses timestamp format 2 (milliseconds). Lyrics use content type 1, markers use content type 4 (events), and chords use content type 5 (chord). Text uses ISO-8859-1 to maximize compatibility with legacy karaoke hardware; unsupported characters are replaced deterministically.

Lyrics

When syllable timestamps are available they are exported; otherwise word timestamps are used, with line timestamps as the final fallback. A newline is appended to the last synchronized token of each line. A USLT frame is also written with the complete unsynchronized lyrics.

Chords

Active chords are ordered by timestamp and stored in a dedicated SYLT frame using the chord content type. Deleted or excluded chords are not exported. The chord symbol is the synchronized cue text.

Markers

Active markers are exported as synchronized events in a dedicated SYLT frame. Deleted or disabled markers are omitted.

Descriptive metadata

TIT2 stores the title, TPE1 the performer/artist, and TCOM the authors/composers when available. A TXXX frame identifies the MTA Audio Editor export profile.

Audio parameters

The profile uses the same MP3 renderer as normal project export. Sample rate 44.1/48 kHz, bitrate 128/192/256/320 kbps and optional normalization are available. A 96 kHz project is resampled to 44.1 or 48 kHz for MP3 export according to the selected output setting. Karaoke metadata is inserted after audio rendering.

Validation

A conforming file starts with ID3 version 2.3, contains at least one lyrics SYLT when synchronized lyrics exist, and contains a SYLT content type 5 when chords exist. Timestamps must be monotonic within each frame. A valid MP3 audio stream follows the ID3 tag.

Limits and interoperability

This document specifies the profile generated by MTA Audio Editor and is not an official manufacturer specification. Different firmware versions may apply different presentation rules for chords, markers, extended characters or colors. Live-production workflows should validate the output on the target model and firmware.

Binary layout / Layout binario

```
ID3v2.3 header
  TIT2 / TPE1 / TCOM
  USLT
  SYLT (lyrics, type=1, ms)
  SYLT (chords, type=5, ms)
  SYLT (markers, type=4, ms)
  TXXX profile id
MPEG Layer III audio frames...
```